

Predicting First-in-Human Pharmacokinetics: Comparative Evaluation of Standard PBPK, High-Throughput PBPK, and Machine Learning

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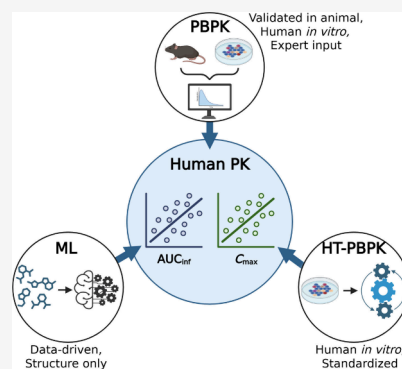
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ABSTRACT: Accurate prediction of human pharmacokinetics is essential for selecting safe starting doses in first-in-human (FiH) trials. This study compares the predictive accuracy of three methodologies including standard physiologically based pharmacokinetic (PBPK) modeling incorporating animal verification, high-throughput (HT) PBPK based on *in vitro* inputs, and machine learning (ML). We evaluated their performance on 40 diverse small molecules that entered clinical development at Roche between 2003 and 2024. Standard PBPK predictions were assessed prospectively from original reports, while HT-PBPK and ML predictions were obtained retrospectively. Predicted area under the curve (AUC_{inf}) and maximum plasma concentration (C_{max}) for oral administration were compared to observed clinical data. While all three approaches demonstrated a useful level of accuracy with at least 49% of predicted AUC_{inf} and C_{max} values within 2-fold, standard PBPK was the most precise method with 65 and 63% of AUC_{inf} and C_{max} within that margin, respectively. This higher precision likely results from the integration of expert judgment and animal *in vivo* data for model refinement, whereas HT-PBPK and ML provide valuable animal-free, high-throughput alternatives. Ensemble models leveraging two or three models were explored, which generally improved accuracy in comparison to single models. This highlights the complementary nature of mechanistic and ML-based models. Currently, standard PBPK is still the preferred method for FiH dose selection for which highest accuracy and mechanistic insight is sought. However, although less accurate, we believe that HT-PBPK and ML are viable animal-free alternatives with value for early human dose prediction, reducing costs and cycle times.

KEYWORDS: Drug Discovery, Human Pharmacokinetics, First-in-Human Predictions, Physiologically based Pharmacokinetic Modeling, Machine Learning



INTRODUCTION

Accurate estimation of the starting dose for first-in-human (FiH) studies is a major contribution of preclinical drug research and an essential first step in clinical drug development. A too high starting dose may expose human subjects to unnecessary safety risks while a starting dose which is too low will result in excessive dose escalations and will be costly and time-consuming.¹ A review of practices at Hoffmann-La Roche at the end of the 1990s² found that empirical dose selection, based on the no observable adverse effect level (NOAEL) in preclinical toxicology studies, was still commonly used. However, this was increasingly superseded by more systematic and theory-based approaches which considered potential pharmacokinetic (PK) differences between human and preclinical species.^{3,4} Human PK was often predicted using allometric scaling. However, for small molecules we started to implement physiologically based pharmacokinetic (PBPK) modeling leveraging *in vitro* data around 2002^{5–7} and using preclinical *in vitro* and *in vivo* data for model verification. An

evaluation of PBPK completed in 2003 demonstrated superior accuracy to allometry for the prediction of human PK⁸ and so thereafter we applied PBPK systematically to predict PK profiles for small molecules entering the clinic. This strategy includes the verification of PBPK predicted PK in preclinical species prior to the final human PK prediction. In cases where this fails in animals due to poor *in vitro* to *in vivo* correlation (IVIVC), it allows for the introduction of empirical scaling factors. While these empirical factors are not mechanistic, they represent a pragmatic approach to acknowledge limitations in *in vitro* assays, current *in vitro* to *in vivo* extrapolation (IVIVE) methods and mechanistic bottom-up models. Their use allows

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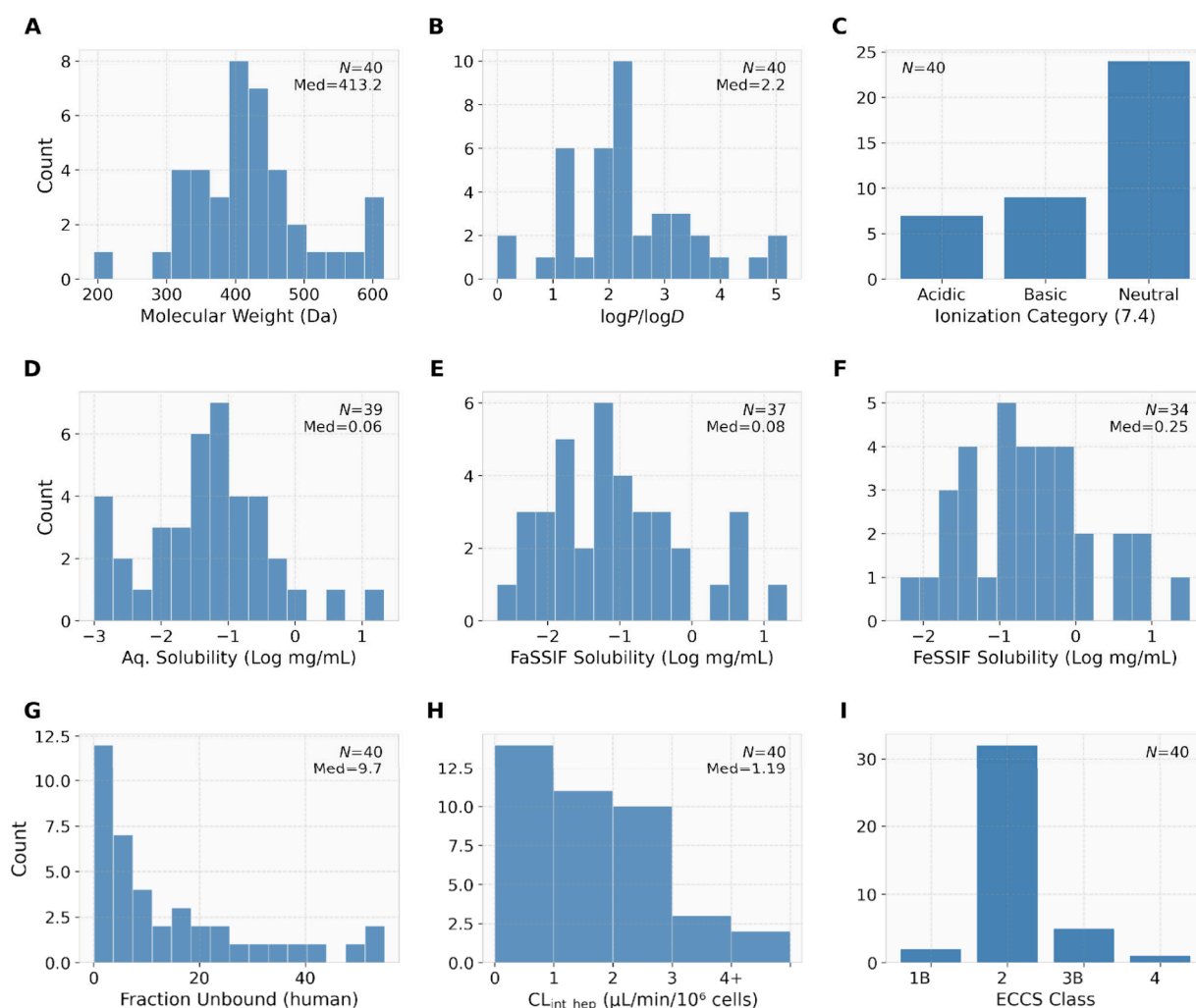


Figure 1. Distribution of molecular properties for the 40 study compounds. The histograms show the distribution of molecular weight (A), lipophilicity (B; logP or logD at pH = 7.4), ionization category (C), solubility in aqueous (D) and biorelevant media (FaSSiF (E) and FeSSiF (F)), fraction unbound in plasma (G), intrinsic clearance (H; CL_{int, hep}), and ECCS classification (I). Insets indicate the number of compounds (N) with available experimental data for each parameter and the median value (Med). Where experimental data was missing for modeling, predictions were made using internal ML models (aq. solubility) or ADMET Predictor (v11.0; Simulations Plus, Lancaster, CA, used for FaSSiF and FeSSiF solubility). Detailed information for each compound is provided in Tables S1 and S2 in the Supporting Information.

PK predictions to incorporate available preclinical *in vivo* data and rely on the scientific judgment of the PK scientist/PBPK modeler. While the long-term aim is to minimize reliance on empirical factors through improvements in IVIVE methodologies, their use has been successfully adopted across pharmaceutical companies.^{9,10}

PBPK models have most commonly been used on a compound-by-compound basis, which limits their use in early discovery due to the expertise required to navigate complex user interfaces and the time-intensive nature of manual data input. Furthermore, the predictive accuracy of PBPK models depends heavily on the quality of input data, which is determined by the reliability of experimental measurements and the selection of the most predictive assays and assay conditions. In early development, such data are often sparse due to limited experimental characterization which would inevitably introduce uncertainty into PBPK predictions. To circumvent these limitations, there is a growing interest in High-Throughput PBPK (HT-PBPK) tools. HT-PBPK consists of automated platforms designed to simplify batch processing for multiple compounds by linking directly to

internal databases containing the relevant human input data. This allows rapid and scalable PK simulations at early stage discovery.^{11–13} At the same time, machine learning (ML) based prediction of PK properties^{13,14} has gained significant interest recently, both preclinically^{15,16} and clinically.^{17,18} ML allows the direct prediction of PK from chemical structure alone and can therefore be used at the early stages without synthesized compounds available. It is, however, limited by the training data and does not replace mechanistic understanding. While the development of PBPK, HT-PBPK, and ML approaches for PK prediction and their application has been comprehensively documented and reviewed,¹³ a head-to-head comparison of their performances using a diverse set of clinical molecules is currently missing.

This study describes the direct comparison of three approaches to predict the human PK for 40 diverse small molecules intended for oral administration which entered clinical development at Roche between 2003 and 2024. We compare PK predictions of the human area under the concentration–time curve extrapolated to infinity (AUC_{inf}) and maximum plasma concentration (C_{max}) to the clinical data

obtained in the FiH studies using three methodologies: Standard PBPK, HT-PBPK, and ML. The standard PBPK approach, incorporating preclinical validation, was applied prospectively at the time of compound progression using only information available prior to the FiH study.¹⁹ In contrast, both the HT-PBPK and ML-based approaches were applied retrospectively. This setup provides a balanced and realistic data-driven evaluation of the translational performance, strengths, and limitations of these methodologies in early clinical development.

MATERIALS AND METHODS

Compound Characteristics

A total of 40 small molecule drug candidates were included in this analysis. They cover multiple therapeutic areas and are intended for oral administration, with prodrugs explicitly excluded. For each compound, comprehensive proprietary physicochemical and *in vitro* ADME (absorption, distribution, metabolism, and excretion) data were available, along with preclinical PK profiles in both rodent and non rodent species. Figure 1 and Supplementary Table S1 present the distribution of key molecular properties. The compounds cover a molecular weight (Mw) range of approximately 200 to 600 Da (median: 413 Da) and lipophilicity values (logP or logD at pH = 7.4) between 0 and 5. The predominant ionization category is neutral and the solubility is generally low (<0.1 mg/mL) in aqueous and fasted- and fed-state simulated intestinal fluid (FaSSIF and FeSSIF). The distribution of fraction unbound in plasma was skewed toward low values (median: 9.7%), i.e. tightly bound and lipophilic compounds are predominant. The distribution of intrinsic clearances ($CL_{int, hep}$), determined using human hepatocyte assays, indicated that the data set consisted mainly of low clearance, metabolically stable compounds (median: 1.2 μ L/min/ 10^6 cells).

Compounds were categorized according to the Extended Clearance Classification System (ECCS)²⁰ to assess their dominant clearance mechanism. It revealed that the majority (80%) of the compounds fell into class 2 (predominantly cleared through hepatic metabolism), with additional representations for classes 1B, 3B, and 4. Detailed methodology and permeability assays utilized for the ECCS assignment are provided in the Supporting Information.

Clinical Data

Human PK data were extracted from finalized clinical study reports for each compound. All studies involving human subjects were approved by the local independent ethics committees and health authorities, and were conducted according to the provisions of the Declaration of Helsinki. Written informed consent was obtained from each study participant before conducting any protocol-related procedures. Additionally, all animal experiments were conducted in full compliance with local, national and ethical principles. All compounds were evaluated in Phase I FiH studies conducted in adult participants under ICH-GCP conditions. Data were primarily derived from single-dose randomized, placebo-controlled, double-blind studies across disease areas and indications. However, for oncology studies employing multiple-dosing, the AUC_{0-24h} obtained on day 1 was used to represent the single-dose profile. To ensure alignment with the modeling strategies, only oral, single agent administration data were considered. Compounds administered exclusively via the intravenous (IV) route were excluded, as were prodrugs, given that their conversion to the active moiety typically necessitates mechanistic modeling. For each eligible compound, PK parameters including C_{max} and AUC_{inf} were obtained using non-compartmental analysis. A complete summary of all available dose levels and associated PK parameters is provided in the Supplementary Table S3. The median number of dose-levels per compound is 6, with a range from 1 to 8.

PBPK Standard Approach with Preclinical Model Verification

The PBPK modeling strategy used in this study has been described in detail in previous papers^{8,9} and is briefly summarized here. The “Standard PBPK” predictions were performed and documented by numerous PBPK modeling scientists prior to the first clinical studies conducted between 2003 to 2024. In all cases the models were implemented in the GastroPlus²¹ software. The disposition PK model comprises 15 tissues and the simulation of oral absorption relies on the Advanced Compartmental Absorption and Transit (ACAT) model.²² The strategy applies generic physiologically based models for animal species and humans integrating relevant compound specific physicochemical and *in vitro* data for ADME processes. In the model validation stage, simulated PK in preclinical species are compared with *in vivo* measurements to assess whether the major ADME processes are adequately captured. Model adequacy was assessed by experienced PBPK scientists based on agreement between simulated and observed plasma concentration–time profiles and key PK parameters falling within acceptable margins (generally within 2-fold). If the animal simulations were considered sufficiently predictive, a human PK prediction was made using the corresponding human *in vitro* inputs. Otherwise, if animal simulations show a mismatch to observed data, the models are refined by experienced PBPK modeling scientists prior to human prediction. Depending on the properties of the compound these refinements might apply to different model inputs. For example, additional clearance processes may be considered, for instance using plated hepatocyte assays in preference to hepatocyte suspensions for compounds which are substrates of uptake transporters.²³ Poor prediction of volume of distribution may require adjustments to the model through use of permeability limited tissue compartments. Poor absorption may require additional measurements of solubility, for example using biorelevant media to replace aqueous buffer. Mispredicted renal excretion may indicate active secretion or reabsorption. For all ADME processes it can happen that an adequate explanation of model mismatches is not possible by the mechanistic data at hand prior to the prediction in humans. In this case, empirical scaling factors may be applied to adjust the human prediction for mismatches observed in animals. While these empirical corrections are not mechanistically explained by physiological processes and may even be misleading in case of species differences there is evidence that overall prediction accuracy can be improved by such adjustments.¹⁰

For this evaluation the standard PBPK predictions were run for the complete set of dose levels tested in the first clinical study allowing a comparison of predicted to observed PK parameters at each dose level for each compound. Since these prospective predictions were made well in advance of the clinical study, the actual demographics of the phase I study population was not known and the simulations were run for a typical healthy human volunteer of 70 kg body weight. However, the absorption model may have been refined based on data collected for the clinical formulation if that was known at the time of the predictions. In summary, “standard PBPK” represents a preclinical PBPK model-based approach leveraging a wide range of preclinical *in vitro* and *in vivo* data and allowing each individual modeler to use their judgment in the final documented prediction. This analysis focuses exclusively on the most likely scenario for clinical PK as determined by the respective PBPK scientist whereas, in most cases, the report also presents different scenarios listing assumptions made and reflecting the inherent uncertainties associated with bottom-up predictions.

High-Throughput PBPK Modeling (HT-PBPK)

SwiftPK is Roche's proprietary HT-PBPK platform designed to provide project teams with fast, scalable and accurate PBPK simulations.²⁴ Leveraging an integrated architecture that grants access to *in vitro* data as well as ML predictions, the tool uses Spotfire Analyst (version 10.3.2; Cloud Software Group, Fort Lauderdale, FL, USA) as the primary user interface. A Spotfire template connects to an internal research database orchestrated via D360[®] (version 23.0.24; Certara, Radnor, PA) to retrieve input data through application

programming interfaces (APIs). Users also have the opportunity to upload external data sets. Within Spotfire, users specify model settings for simulations, submit the simulations and, once terminated, the outputs are returned to Spotfire for analysis using a set of preconfigured, customizable visualizations. A key capability of SwiftPK is its hierarchical parameter sourcing. When experimental measurements are unavailable for a compound, the system automatically retrieves the corresponding ML-predicted values from our internal Machine Learning Property Prediction (MLPP) platform,²⁵ which is described in the [Supporting Information](#). Because this approach is designed to simulate an early discovery workflow, all predictions are generated strictly bottom-up without any empirical fitting or compound-specific model refinement.

As was done for PBPK, SwiftPK projections were generated at all dose levels investigated clinically. All simulations were performed using the HTPK module of ADMET Predictor (v11.0; Simulations Plus, Lancaster, CA). Since the aim is to predict PK parameters from oral dosing, the module's immediate-release tablet model was employed. The HTPK module incorporates the ACAT model derived from the GastroPlus PBPK platform. For computational efficiency, HTPK simplifies systemic distribution into a single "central" compartment. The volume of this compartment, termed the mechanistic volume of distribution, is nonetheless predicted accounting for tissue composition variations. For all compounds we used a volume of distribution calculated using the Lukacova extension²⁶ of the Rodgers method.^{27,28}

In our study, the *in vivo* clearance was calculated from the *in vitro* hepatocyte intrinsic clearance ($CL_{int, hep}$) using the well-stirred liver model. This calculation incorporated fractions unbound in plasma and *in vitro* binding estimated via the dilution scaling method which estimates *in vitro* binding from the incubation media containing 10% FCS.²⁹

The following *in vitro* inputs (see also [Supplementary Tables S1 and S2](#)) were directly taken from the internal discovery database: plasma protein binding, blood-to-plasma ratios, $CL_{int, hep}$, $\log P$ or $\log D$ at pH = 7.4 depending on availability, aqueous solubility, FaSSIF and FeSSIF solubilities, and drug particle size. Regional *in vivo* human intestinal effective permeability (P_{eff}) was calculated from the *in vitro* apparent permeability (P_{app}). In this study we prioritized the P_{app} measurements coming from the Caco-2 cell system. If not available for a specific compound, then the P_{app} from apical-to-basolateral (AB) permeability in LLC-PK1-Mdr1a cells (porcine kidney epithelial cell line overexpressing mouse P-gp)³⁰ was considered and, if not present, the P_{app} coming from the PAMPA³¹ assay was taken. When the experimental value was missing for aqueous solubility (only 1 compound), the corresponding ML-predicted value coming from MLPP was used instead, whereas for missing blood-to-plasma ratios (4 compounds) the ADMET Predictor (v11) was used. pK_a , ionization categories, and diffusion coefficients were calculated with ADMET Predictor (v11). Finally, standard GastroPlus default parameters of particle density (1.2 g/mL), dose volume (250 mL), and body weight (70 kg) were applied. When measured data on clinical particle radius was not available, a default value of 10 μm was used.

Machine Learning PK Predictions

ML models can predict human PK parameters directly from chemical structure, which would allow their use in very early stages of discovery. However, human PK data sets are small and sparse which often limits the generalizability of models trained on clinical data alone. To address this challenge, our colleagues Komissarov and co-workers developed a multitask (capable of predicting multiple end points simultaneously) graph neural network (GNN) for human PK prediction.³² Since the original work demonstrated promising predictive performance and the model has since been deployed within our internal infrastructure to support early discovery teams, its capabilities are further evaluated in this comparative analysis. Only a high-level methodological overview is given here whereas comprehensive technical details can be found in the original publication.³²

The clinical PK data used for the original model training was curated from Elsevier's Pharmapendium and Clarivate Cortellis Drug Discovery Intelligence. The raw data was preprocessed and filtered to include only healthy adult populations administered single-compound, single-dose regimens via oral or IV routes. Further selection criteria restricted entries to plasma, serum or blood matrices, molecular weight and molar doses to smaller than 1000 g/mol and 0.01 mol, respectively. Outliers were removed using a replicate-based Z-score combined with an elliptic envelope method. The resulting data set contained roughly 2700 unique molecular structures covering 11 PK end points.

Inputs to the model are the 2-dimensional molecular structure and the molar dose, which is included as an additional, scalar feature. The model architecture is based on GraphGPS³³ that was modified by implementing graph convolutions with GATv2 layers³⁴ and processing edge attributes through a multilayer perceptron at each GraphGPS layer. To improve generalizability, the model was trained in a multitask fashion simultaneously on the 11 PK end points and an auxiliary data set of human *in vitro* end points including, e.g., solubility, lipophilicity and membrane permeability for roughly 200000 molecules. To ensure an unbiased evaluation, the 40 compounds studied in this work were cross-checked against the training data and all corresponding *in vitro* and *in vivo* data was explicitly excluded prior to model retraining. After removing exact matches, a similarity analysis against the human PK-annotated training set confirmed the structural independence of the test compounds. The mean maximum Tanimoto similarity (based on extended-connectivity fingerprints (ECFPs)³⁵ with a radius of 2 and 2048 bits) was 0.30, with 88% falling below 0.4 and zero compounds exceeding a threshold of 0.7.

For uncertainty quantification, various approaches were tested in the original publication, but the final model utilized a combination of mean–variance estimation³⁶ and ensembling.³⁷ The obtained uncertainties are then discretized into a 1–10 goodness score, where 1 denotes maximal and 10 minimal uncertainty.

Performance Evaluation

Model performance for PBPK, HT-PBPK and ML predictions were evaluated based on their ability to predict human C_{max} and AUC_{inf} values across all dose levels investigated clinically. Statistical evaluation focused on aspects including accuracy and bias.³² General goodness-of-fit was quantified using the root mean squared error (RMSE), which is calculated based on the Log-transformed human and predicted parameters. To differentiate between bias and precision, we utilized the Average Fold Error (AFE) and Absolute Average Fold Error (AAFE). The AFE serves as a measure of bias, for which values <1 or >1 indicate a tendency toward under- or overprediction, respectively. In contrast, the AAFE quantifies the average magnitude of the error, regardless of direction. It should be noted that the RMSE was included alongside the fold-error metrics because it is more sensitive to large outliers, whereas AAFE reflects the typical error magnitude. The mathematical equations are detailed in the [Supporting Information](#). Finally, the percentage of predictions falling within 2- and 3-fold of observed clinical data was calculated as a standard metric. All analyses were performed using Python 3.12.

RESULTS

Global Analysis

We compared the performance of PBPK, HT-PBPK and ML models for two critical PK parameters in FiH studies, AUC_{inf} and C_{max} . The results are summarized in [Table 1](#). Correlations between predicted versus observed values are presented in [Figures 2](#) (AUC_{inf}) and [3](#) (C_{max}). The data set included 40 compounds with 229 to 239 observations across various doses for AUC_{inf} and C_{max} , respectively. The accuracy of the predictions, as measured by logRMSE, ranged from 0.38 to 0.54 across all methods. Standard PBPK achieves the lowest errors in predicting the clinical observations with logRMSEs of

Table 1. Statistical Performance Metrics for PBPK, HT-PBPK, and ML Predictions^a

	AUC _{inf}			C _{max}		
	PBPK	HT-PBPK	ML	PBPK	HT-PBPK	ML
logRMSE	0.38	0.54	0.47	0.40	0.43	0.47
AFE	0.93	0.93	1.05	0.91	0.93	1.05
AAFE	1.32	1.52	1.42	1.34	1.42	1.40
2-fold (%)	64.63	48.91	53.28	62.50	48.75	57.08
3-fold (%)	81.66	64.19	71.62	77.50	77.01	77.50

^aEvaluation metrics include the Root Mean Square Error (logRMSE), Average Fold Error (AFE), Absolute Average Fold Error (AAFE), and the percentage of predictions falling within 2-fold and 3-fold error margins. logRMSE refers to a quantity calculated based on log transformed human and predicted values.

0.38 and 0.40 for AUC_{inf} and C_{max}, respectively. While HT-PBPK showed the largest average errors for AUC_{inf} being ~50% larger than for PBPK, it was close to the PBPK performance for C_{max} (~5% deviation). For ML, the error magnitude was constant at ~0.47 for both AUC_{inf} and C_{max}.

The AFE indicates that the predictions exhibited a slight systematic bias across all three approaches. AFE values remained close to unity (ranging from 0.93 to 1.05 for AUC_{inf} and from 0.91 to 1.05 for C_{max}). With the exception of ML, these values are consistently below 1.0, which reflects a general tendency to underpredict exposure. The AAFE ranged from 1.32 to 1.52 for AUC_{inf} and from 1.34 to 1.42 for C_{max}. This translates to an average deviation of roughly 30% to 40% from observed values. The HT-PBPK and ML approaches showed slightly higher AAFE values (1.40–1.52) in comparison to PBPK, which is consistent with the wider dispersion as quantified by the logRMSE values.

Overall, all three methodologies predicted the majority of the observed parameters within 2- or 3-fold error margins (Table 1, Figure 2 and Figure 3). For standard PBPK, 65% of AUC_{inf} and 63% of C_{max} values fell within 2-fold of the observed human PK data. When evaluated using the 3-fold

error criterion, these percentages increased to approximately 82% and 78%, respectively. While HT-PBPK predicted roughly half of the data within 2-fold for both parameters, the performance improved to 64% for AUC_{inf} and 77% for C_{max} at the 3-fold level. The ML model correctly predicted 53% (AUC_{inf}) and 57% (C_{max}) of values within 2-fold, which increased to 72% and 78% within the 3-fold boundary.

Compound-Level Analysis

Besides across-study comparisons, the performance of the three methods was also scrutinized for individual compounds. Figures 4 and 5 visualize the log transformed prediction errors for AUC_{inf} and C_{max} at the individual compound and dose levels. The vertical lines connecting the data points for single compounds highlight dose-dependent trends in prediction accuracy. Despite comparable global error metrics (see Table 1), compound-specific analysis reveals notable outliers and intriguing trends. For example, RO14 was consistently overpredicted for both PK parameters by all three methodologies, with errors exceeding the 3-fold margin across most dose levels. Similarly, large dose-dependent variations in prediction errors were observed for compounds such as RO35, RO34, and RO29. Furthermore, a direct comparison of the mechanistic approaches revealed instances where the HT-PBPK workflow struggled relative to the standard approach. For example, the accuracy of HT-PBPK for RO5, RO6, and RO15 (all of which belong to ECCS 2) dropped notably in predicting AUC_{inf} as well as for RO15 when predicting C_{max}. In the case of RO15, the HT-PBPK method significantly underpredicted parameters (>3-fold error), whereas standard PBPK remained largely within 2-fold boundaries. Similarly, the ML model performed poorly for certain compounds, such as RO20 or RO24, for which it failed to match the accuracy of the PBPK methodologies. Here, both the standard and HT-PBPK methods provided predictions largely within the 2-fold range, whereas the ML model significantly overpredicted both parameters. Yet there are cases, such as RO31 and RO40, for which the ML approach outperforms the PBPK based methodologies in predicting AUC_{inf}. Conversely, RO8 was a

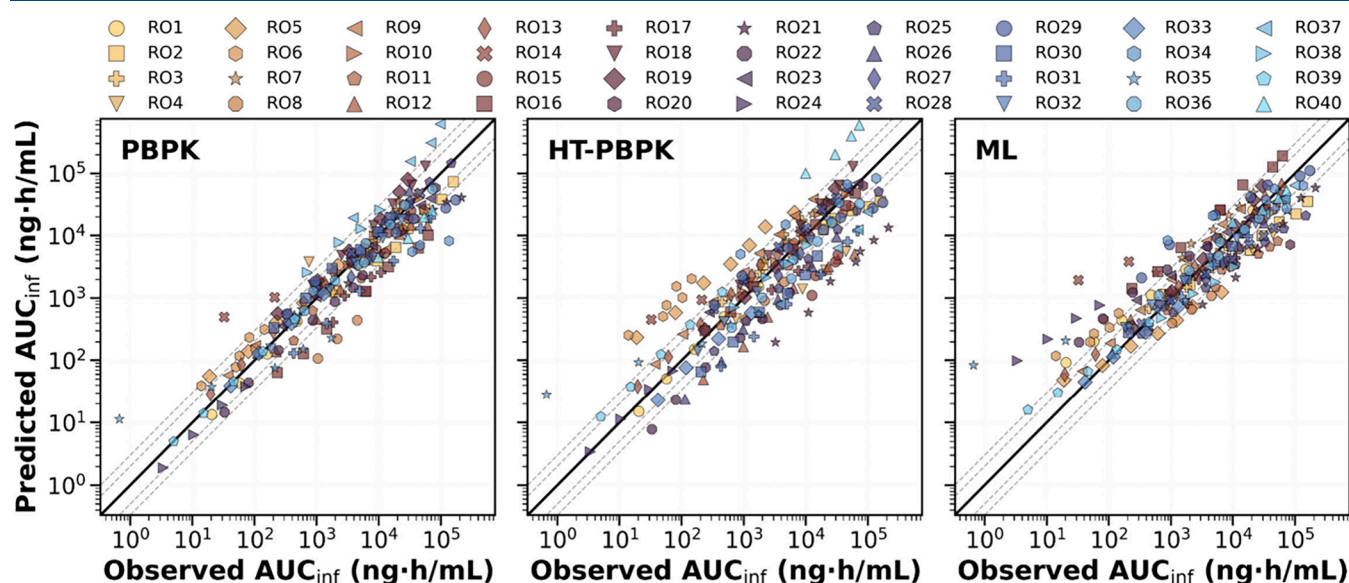


Figure 2. Performance of PBPK, HT-PBPK and ML for predicting AUC_{inf} following oral administration at different doses. Symbols represent individual compounds (*N* = 40) plotted on a log scale. The solid black line represents the line of unity while the inner and outer dashed lines represent the 2-fold and 3-fold error boundaries, respectively.

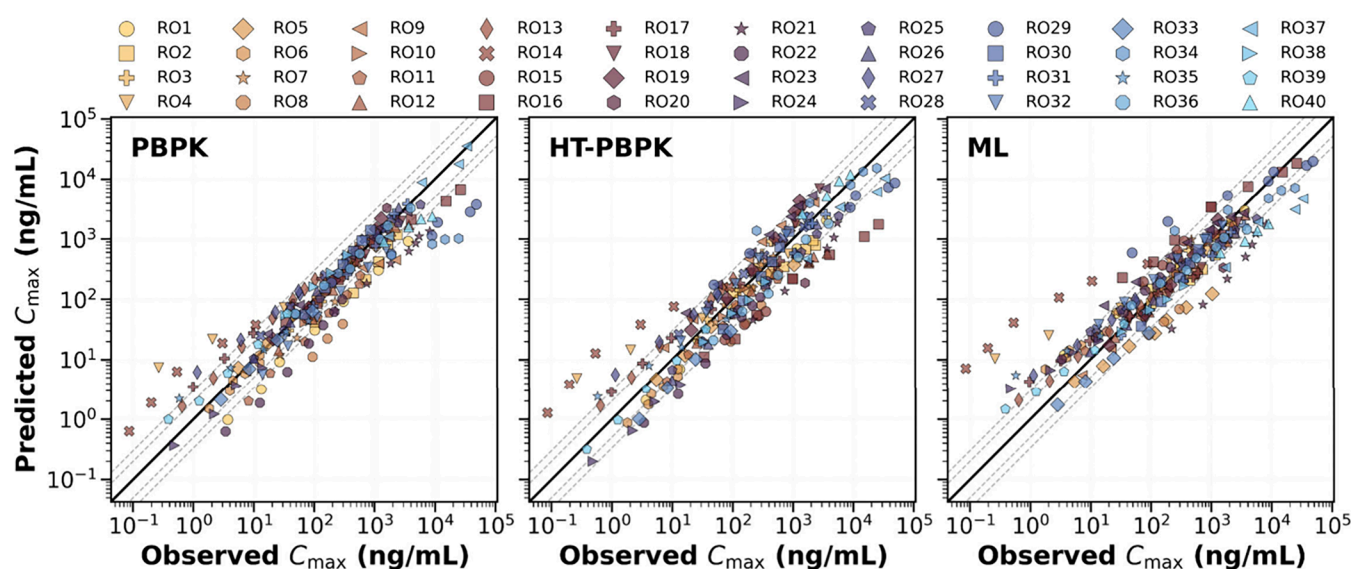


Figure 3. Performance of PBPK, HT-PBPK and ML for predicting $C_{\max}$ following oral administration at different doses. Symbols represent individual compounds ($N = 40$) plotted on a log scale. The solid black line represents the line of unity while the inner and outer dashed lines represent the 2-fold and 3-fold error boundaries, respectively.

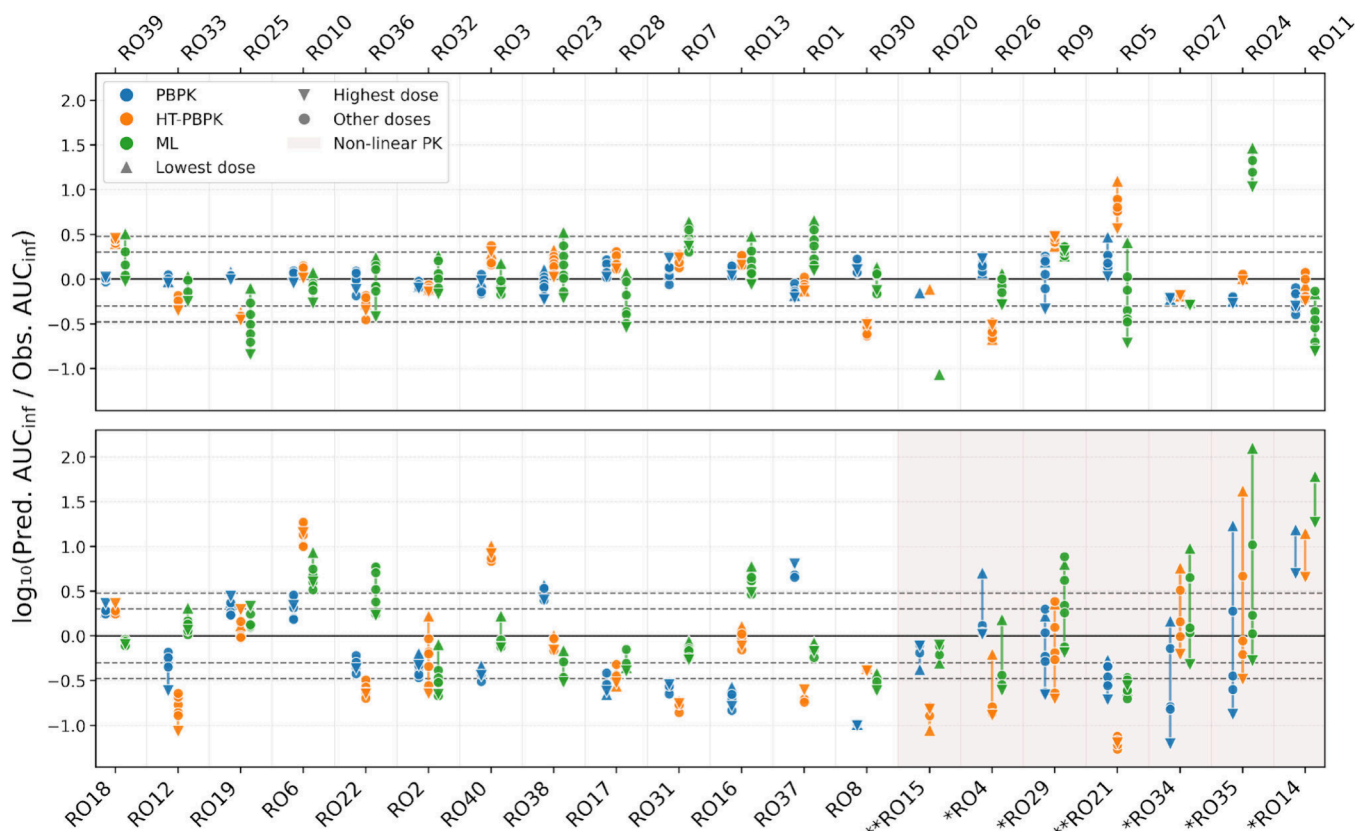


Figure 4. Log transformed prediction errors for human AUC_{inf} are shown for all compounds across three modeling approaches: PBPK (blue), HT-PBPK (orange), and ML (green). Compounds are ordered by increasing PBPK prediction error, with linear PK compounds shown first (white background) followed by nonlinear PK compounds (shaded background). For each compound, individual dose levels are displayed as separate markers, with upward triangles indicating the lowest dose, downward triangles indicating the highest dose, and circles representing intermediate doses. Vertical lines connect measurements across the dose range for each method-compound combination, illustrating dose-dependent trends in prediction accuracy. The horizontal solid line at $y = 0$ represents perfect prediction. Dashed horizontal lines indicate 2-fold and 3-fold error boundaries. Positive values indicate overprediction, while negative values indicate underprediction. The three methods are horizontally offset for visual clarity. Compounds with observed nonlinear PK are marked with * or ** if they show supra- or subproportional behavior, respectively. Linearity classification was based on log–log regression of AUC_{inf} vs dose, where the slope equals 1.0 for perfectly dose-proportional (linear) PK. Compounds for which the slope differed by more than 0.25 from 1.0 were classified as nonlinear.

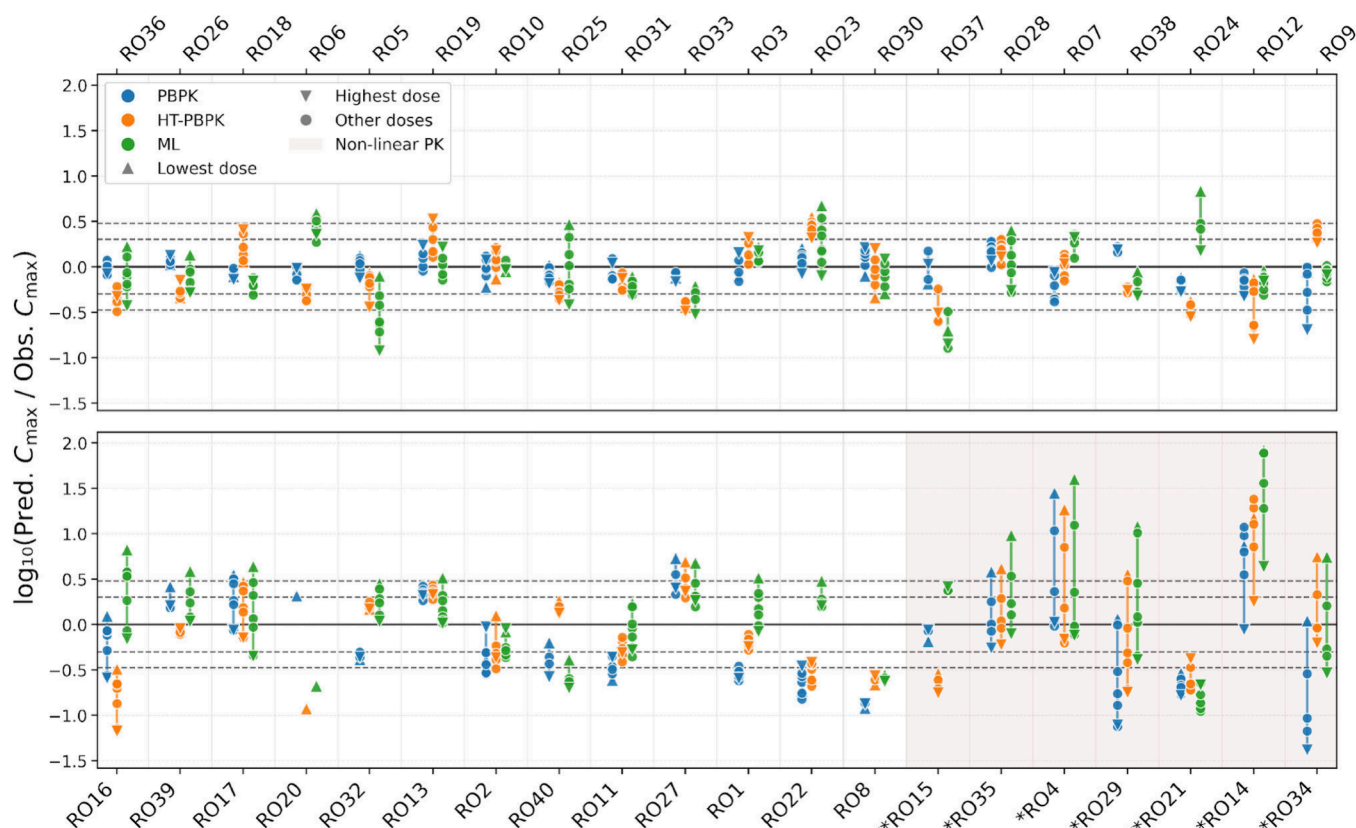


Figure 5. Log transformed prediction errors for human $C_{\max}$ are shown for all compounds across three modeling approaches: PBPK (blue), HT-PBPK (orange), and ML (green). Compounds are ordered by increasing PBPK prediction error, with linear PK compounds shown first (white background) followed by nonlinear PK compounds (shaded background). For each compound, individual dose levels are displayed as separate markers, with upward triangles indicating the lowest dose, downward triangles indicating the highest dose, and circles representing intermediate doses. Vertical lines connect measurements across the dose range for each method-compound combination, illustrating dose-dependent trends in prediction accuracy. The horizontal solid line at $y = 0$ represents perfect prediction. Dashed horizontal lines indicate 2-fold and 3-fold error boundaries. Positive values indicate overprediction, while negative values indicate underprediction. The three methods are horizontally offset for visual clarity. Compounds with observed nonlinear PK are marked with * or ** if they show supra- or subproportional behavior, respectively. Linearity classification was based on log–log regression of AUC_{inf} vs dose, where the slope equals 1.0 for perfectly dose-proportional (linear) PK. Compounds for which the slope differed by more than 0.25 from 1.0 were classified as nonlinear.

rare case where PBPK was significantly less accurate in comparison to HT-PBPK and ML for reasons that are currently unclear.

One reason for compound-specific performance deviation is nonlinear PK. Compounds with nonlinear PK are explicitly marked in Figures 4 and 5 to highlight their impact of dose-dependency on performance. Specifically, symbols denote supra-linear (*, exposure increasing more than proportionally) and sublinear behavior (**, exposure increasing less than proportionally). These compounds often exhibit larger deviations from observed values. This is particularly evident in the supra-linear (*) cases, where we observe substantial dose-dependent changes in the magnitude of the prediction error.

Machine Learning Confidence

In addition to the predictions of the PK parameters, the ML model reports its confidence for each prediction. Hence it was examined whether the goodness score could distinguish between accurate predictions and outliers. As illustrated in Figure 6, there is a strong correlation between the estimated model confidence and the observed accuracy for both AUC_{inf} and $C_{\max}$. High-confidence predictions (blue) cluster tightly around the line of identity, while predictions flagged with

higher uncertainty (red) correspond to larger deviations from observed values. This can be analyzed quantitatively by calculating the error metrics for high (>5) and low goodness (≤ 5) groups, see Table S4. For both AUC_{inf} and $C_{\max}$ the high goodness group showed a reduction in logRMSE and a substantial increase in the proportion of predictions falling within 2-fold of observed values. Significant deviations, as mentioned above for RO24, were predominantly associated with lower goodness scores (indicated in red and orange).

DISCUSSION

PBPK is well established as the preferred approach for predicting FiH pharmacokinetics, offering a mechanistic framework to integrate preclinical *in vitro* and *in vivo* data for human PK predictions.⁹ However, it requires substantial time and resources to generate input data and strong expertise to perform the manual modeling for individual compounds. There has been a strong interest in developing HT-PBPK approaches which streamline data management and simplify complicated user interfaces.¹¹ Furthermore, the demonstrated power of ML models in multiple fields has motivated efforts to apply this technique for the prediction of pharmacokinetics from chemical structure alone.^{17,18,32} This work provides a comprehensive evaluation of these three predictive method-

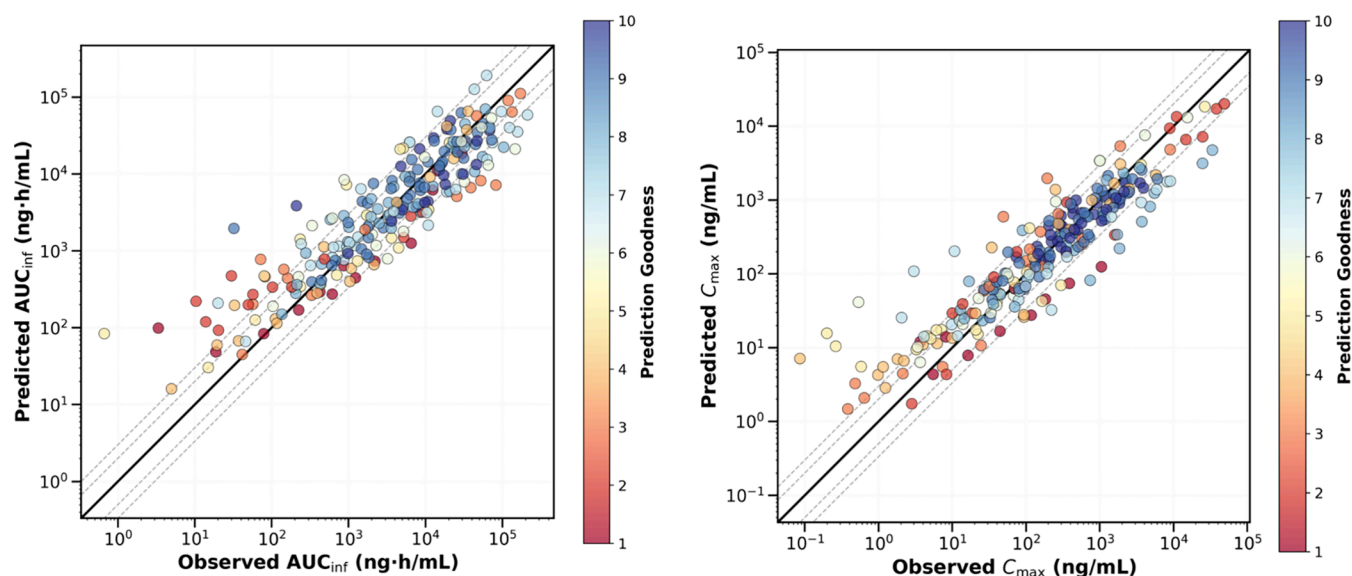


Figure 6. Machine learning predictions for the same compounds as shown in Figures 2 and 3. The ML predictions are colored according to their goodness, which illustrate high confidence and low confidence predictions in blue and red, respectively. Lower goodness scores (scores <5, indicated in red and orange) can arise when the test compound is structurally dissimilar to the training set or, as dose is a direct model input, stem from atypical dose levels. Additionally, a low confidence can be caused by the output space itself if the predicted AUC_{inf} or C_{max} values fall outside the boundaries established by the training data distributions. The latter can be analyzed by comparing the predictions to the 10%/90% quantile of the training data, which are roughly 40/56000 ng·h/mL and 4/4000 ng/mL for AUC_{inf} and C_{max} , respectively. Hence, comparing these boundaries to the figure, the magnitude of the PK parameter is likely to have a major impact on the uncertainties.

ologies for clinical candidates. It aims to assess their accuracy in predicting human PK parameters via translation of preclinical data or directly from structure using a diverse set of compounds investigated clinically following oral administration.

Overall, all three methods delivered a useful level of predictivity with at least 49% of predicted AUC_{inf} and C_{max} values within 2-fold of the observed value and at least 64% within 3-fold. Standard PBPK was the most precise method with 65% and 63% of AUC_{inf} and C_{max} predictions within the 2-fold margin. This can be compared to the study by Mao et al.¹⁰ which described PBPK incorporating animal-derived scaling factors and demonstrated high accuracy with 94% of C_{max} and 72% of AUC_{inf} predictions within a 2-fold error. We believe that this high level of accuracy for standard PBPK compared to HT-PBPK results from additional expert judgment and model refinement. HT-PBPK represents a standardized modeling approach using fixed *in vitro* to *in vivo* scaling approaches which are applied for all compounds and without necessitating any consideration of IVIVE verification using animal *in vivo* data. In contrast, standard PBPK described here (similar to the approach described by Mao et al.) allows the modeling scientist to use their judgment and experience to refine the human model prior to the final prediction. In this study, for AUC_{inf} , standard PBPK reduced the logRMSE by more than 30% compared to the HT-PBPK workflow (0.38 vs 0.54) although for C_{max} the improvement was less significant. This level of improvement might be considered important, but it does require significantly more expert input, time and resources whereas HT-PBPK can be largely automated and can therefore be run for many more compounds. Furthermore, it is important to note that HT-PBPK does not require the use of animal data which is an important consideration in view of the 3R's (replace, reduce, refine) of animal studies in pharma research.³⁸ Other studies have prioritized animal-free HT

screening for earlier development stages. Miljković and co-workers¹⁷ used ML to predict ~ 50 and 60% of AUC_{inf} and C_{max} , respectively, within 3-fold. Geci et al.¹² used HT-PBPK to predict AUC_{inf} and C_{max} within a 10-fold error for 84 and 87% of the studied compounds.

Since our analysis used comprehensive, late-stage *in vitro* packages, the findings for HT-PBPK likely reflect an upper bound of predictive accuracy. Such a level of experimental characterization is typically not available earlier, e.g. lead optimization, which makes the HT-PBPK results somewhat optimistic for application at that stage. To simulate this early stage scenario, the HT-PBPK simulations were rerun with ML-predicted inputs only. It should be noted, however, that potentially some compounds were part of the training set for the ML models used to predict the inputs. The results are shown in Figure S1. They confirm a drop in performance in particular for AUC_{inf} , for which a systematic underprediction is visible (AFE = 0.67). The percentage of predictions falling within 2-fold of the observed values are 38.5 and 49.2% for AUC_{inf} and C_{max} , respectively. Nevertheless, we believe HT-PBPK remains a viable strategy even when data is sparse. A recent study on rodent PK predictions showed that replacing all *in vitro* inputs with ML predictions still results in errors of less than 4-fold for most compounds.²⁴ That work also demonstrated that you can improve performance by only considering compounds for which the ML goodness score is high. Additionally, the study found that ECCS2 compounds (predominantly cleared through hepatic metabolism) were better predicted, whereas ECCS3 and ECCS4 compounds showed lower prediction accuracy. For the present data set of 40 compounds, a statistical evaluation of the influence of ECCS class on accuracy was less conclusive due to the data set imbalance, as the majority of compounds (32 out of 40) belong to class 2. Although a decrease in accuracy is typically expected for nonclass 2 compounds, the present data did not show a severe

drop in performance across the remaining classes. However, given that Classes 1B, 3B, and 4 contain only 2, 5, and 1 compounds, respectively, the small sample sizes prevent drawing reliable statistical conclusions. Nevertheless, the correlations and performance metrics across ECCS classes are provided for completeness in Figure S2 and S3, and Tables S5 and S6.

A major limitation in modeling human PK is the prediction of dose nonlinearity. For supra-linear compounds, significant dose-dependent variations in error were visible. Hence, the current models failed to capture the underlying nonlinear mechanisms which reflect the lack of inputs on saturation of metabolic enzymes or transporters. For the ML model, this limitation is likely caused by the scarcity of nonlinear training examples. In contrast, all methodologies were more robust in capturing sublinear trends. The comparable error magnitudes across dose levels for sublinear cases suggest that limiting factors such as saturable absorption or solubility constraints are better represented. This is supported by the low measured solubility for compounds such as RO15 (0.001 mg/mL) and RO21 (0.009 mg/mL). A detailed discussion of individual compound predictions is out of scope of this work. It is, however, interesting to explore a few examples which were predicted less well and where subsequent clinical data and analysis have highlighted gaps.

Selected Case Studies

RO14 showed >3-fold overpredicted $C_{\max}$ and AUC_{inf} for all methods (Figures 4 and 5). It is a low Mw compound (194 Da) with a strong basic group with high solubility and high permeability (BCS1). Renal elimination was minor. In monkey and dog, the scaled hepatic clearance was in good agreement with the *in vivo* total blood clearance but was ~ 3-fold underpredicted in the rat. The final prediction used scaling from human hepatocyte suspension intrinsic clearance to estimate *in vivo* clearance as 12 mL/min/kg and complete absorption was simulated leading to overall oral bioavailability of 48%. Subsequently, modeling of the clinical pharmacokinetic data indicated that hepatic clearance had been underestimated, and since the predicted clearance was already ~ 50% of liver blood flow the hepatic first pass extraction was greatly underestimated resulting in a lower bioavailability than predicted. Further clinical and *in vitro* explorations indicated that glucuronidation by the UGT2B10 isoform played a significant role in human metabolism³⁹ and this enzyme was not fully active in the *in vitro* incubations. Intriguingly, the ML predictions were even worse than the mechanistic modeling for RO14, with strongly overconfident goodness scores particularly for AUC_{inf} (see Figure 6).

RO35 is a neutral, BCS1 molecule with a Mw of 431 Da. *In vitro* to *in vivo* scaling of hepatic clearance from hepatocytes could be verified in preclinical species and measurements in human hepatocytes consistently predicted a low clearance with no concentration dependency. Therefore, the estimate of human hepatic clearance was confidently expected to be low (in the range of 2 to 5 mL/min/kg). Distribution volume was larger than total body water and reasonably predicted in preclinical species based on physicochemical properties. In humans the V_{ss} was expected to be around 1.4 L/kg. However, RO35 showed >3-fold over predicted AUC_{inf} and $C_{\max}$ at the lowest doses with the prediction accuracy improving as the dose increased. Subsequent clinical data and modeling showed that mispredictions were related to the very low starting dose

of 0.3 mg and the influence of target mediated drug disposition (TMDD), which became saturated as dose increased and resulted in better prediction at higher dose. The widespread of predicted/observed ratio for different dose levels with both mechanistic and ML approaches (Figures 4 and 5) in this case is due to the predicted linear PK, whereas strong dose nonlinearity was observed. However, as with PBPK, the ML prediction improved with increasing doses with AUC_{inf} and $C_{\max}$ falling within the 2-fold margin. This is also reflected in the corresponding goodness scores which increased from 5 to 9. It is hypothesized that the lower confidence arises when predicted values fall outside the boundaries established by the training data distributions. TMDD has been reported for other small molecule drugs⁴⁰ and is recognized as being difficult to predict preclinically. RO4, a monoamine oxidase inhibitor which was reported by us in a previous publication,⁴⁰ is another example of this phenomenon. These observations indicate that TMDD is one common cause of mismatches between predictions and observations.

Operational Considerations and Future Perspectives

The three methodologies differ fundamentally in their operational demands and data requirements. Standard PBPK relies heavily on expert judgment to refine models and provides a mechanistic basis for regulatory submission. Both standard and HT-PBPK necessitate (though to different extents) the availability of comprehensive compound-specific *in vitro* ADME data, with standard PBPK also requiring preclinical *in vivo* PK profiles, while ML requires a structure only. Depending on the project phase (e.g., lead identification vs optimization), these lower-requirement methods provide the necessary throughput to rank compounds and drive early prioritization based on PK liability. Hence, these approaches should be viewed as complementary rather than competitive. In this comparative analysis, standard PBPK emerged as the most accurate approach. We believe that this is largely due to the human in the loop and preclinical *in vivo* data. For example, the more accurate prediction of RO15 by PBPK relied on a rat-derived scaling factor to correct the clearance scaling for high protein binding, which cannot be captured by HT-PBPK (see compound 1 in ref 9). However, this reliance on animal data raises an important question for the future of drug development as it is in contrast to industry initiatives toward new approach methodologies and the reduction of animal testing. Hence, although PBPK offers the highest precision for regulatory dose predictions, it is essential to evaluate whether the improvement in accuracy over standardized *in vitro* only approaches always justifies the use of animal-derived data. Looking to the future, the adoption of advanced *in vitro* tools such as HepatoPac^{41–43} or organ-on-chip systems⁴⁴ offer promising alternatives to close this gap and reduce reliance on animal studies.

Both HT-PBPK and ML offer a scalable, ethical alternative for early stage chemical prioritization and safety screening. In this study, the ML model showed good performance even in the absence of explicit mechanistic input. Despite these promising results, ML is currently rarely utilized for human PK prediction. This may be in part due to skepticism regarding its “black box” nature and may also be because of the lack of mechanistic insight which it provides. Furthermore, the 40 compounds analyzed here are inherently biased toward compounds that successfully reached human *in vivo* testing stages, which may not reflect the broader chemical space

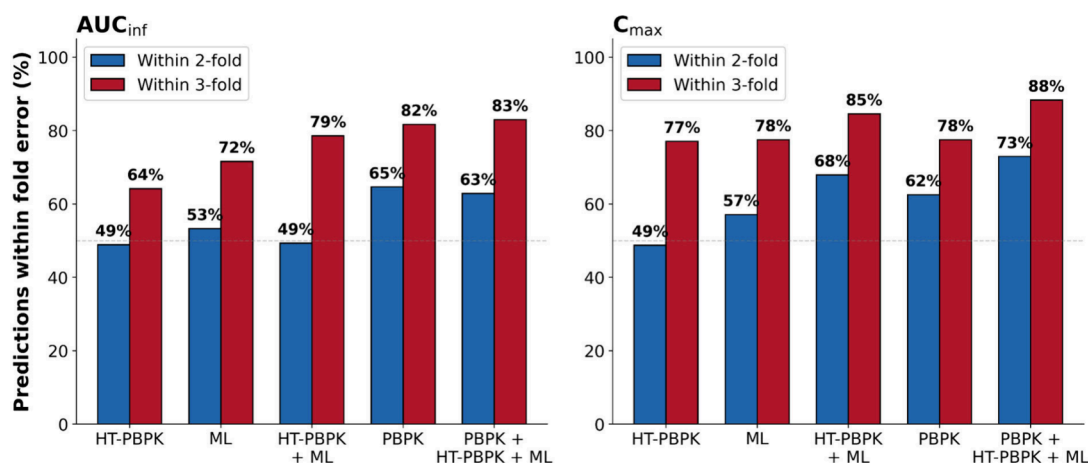


Figure 7. Impact of ensemble modeling on prediction accuracy. The bar charts display the percentage of predictions that are within 2-fold (blue) and 3-fold (red) error margins for AUC_{inf} (left) and C_{max} (right). The analysis compares individual methodologies (HT-PBPK, ML, and Standard PBPK) against ensemble approaches formed by averaging the predictions of the constituent methods.

encountered during early design. The performance of these models on more diverse sets, where ML and HT-PBPK methods would be most valuable, is nearly impossible to be established (only compounds that successfully progressed through development reach clinical testing which inevitably retains the “survivor bias”). The ML model benefits from the accumulation of historical human PK and auxiliary physicochemical data over time. Evaluating the model in a strict “simulated prospective scenario” (e.g., predicting a compound from 2003 using only pre-2003 training data) would severely restrict the training data set and yield overly pessimistic estimates for today’s application. This is balanced by ensuring the test set is structurally different from the PK training data (maximum Tanimoto similarity for the test compounds is below 0.7, with most falling below 0.4), as is frequently done for ML studies. Nevertheless, the current findings suggest that ML offers a level of accuracy that should be integrated into testing cascades, e.g. alongside HT-PBPK. For example, one way to further enhance reliability, is to implement an ensemble approach (averaging the predictions of different “expert systems”) to smooth out individual model biases and provide a more robust estimate of human pharmacokinetics. This is demonstrated to be a promising approach as is shown in Figure 7, which yields further improvement in particular for C_{max}. This finding reinforces the idea that these methodologies are not mutually exclusive but synergistic. With continued advancements in ML technology and growing training data sets, the synergistic use of ML and PBPK methodologies holds significant potential to improve predictive modeling in drug development.

CONCLUSION

This study provides the first comprehensive evaluation of standard PBPK, HT-PBPK, and ML approaches for predicting human pharmacokinetics using diverse internal clinical data. While standard PBPK consistently delivered the most precise predictions, all three methodologies demonstrated a useful level of accuracy. Ensemble methods generally achieved superior performance over individual models. We suggest that these approaches are complementary with both model philosophy (mechanistic understanding and prediction optimization) and performance, and that ensemble strategies may further enhance prediction reliability. Additionally, addressing

the prediction of nondose linear kinetics remains an area of opportunity for advancing translational science.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.molpharmaceut.6c00429>.

Tables on molecular and physicochemical properties (Tables S1 and S2), a comprehensive overview of the observed and predicted PK parameters (Table S3). Additionally, the ECCS system and the performance metrics are described. Additional evaluations contain performance metrics for the ML predictions for low-goodness (≤ 5) and high-goodness (> 5) subgroups (Table S4), performance of HT-PBPK with ML inputs (Figure S1) and an analysis of the performance of the three methods based on the compound ECCS classes (Figure S2 and S3; Tables S5 and S6) (PDF)

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Notes

The authors declare the following competing financial interest(s): All authors are employees of F. Hoffmann-La Roche Ltd. The authors have no other relevant affiliations or financial involvement with any organization or entity having a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript. F. Hoffmann-La Roche Ltd. holds paid licenses for Certara and Simulations Plus software. All analyses and subsequent interpretations were performed independently by Roche.

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